BABU BANARASI DAS UNIVERSITY

SCHOOL OF COMPUTER APPLICATION
BCACS21

PROJECT OF:- APPLIED END TO END SECURITY TO CLOUD APPLICATIONS (IBM)

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Project Section - 1

Health care DATA Security

CASE Study :- Healthcare Data System

1. introduction :- Hospitals stores sensitive data.
   - includes
     - Name, age, address
     - Medical history
     - Reports & filling information

   Data must be secure & private

2. Data Analysis & Trends :-
   - Used to
     - Identify desease patterns
     - Track patient records
     - Improve treatment

   Risk - Data exposure during analysis

3. Possible Risks :-
   - Data breach (hackers)
   - Insider misuse (staff)
   - Data leakage
   - Ransomeware attacks
   - Legal / compilance issues

4. Data loss Prevention (DLP) :-
   → Prevents unauthorized data sharing

   Functions :-
   - monitors emails & file transfers

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- Blocks sensitive data sharing
- Sends alerts

Example :- If patient data is emailed outside → Blocked automatically.

5. <u>Data Anonymization</u> :-

→ Hides patient identity

<u>Methods</u> :

- Masking → Name → Patient ID
- Encryption → Data unreadable
- Tokenization → Replace data with token
- Aggregation → use grouped data

Example :- Rahul , Age - 45 , Diabtes
Patient Id 567, age group 40-50, Diabetes

6. <u>Security Solutions</u> :

→ Role-based access (RBAC)
- Multi Factor authentication (MFA)
- Data encryption
- Regular Audits
- Backup system.

7. <u>Conclusion</u> :-

By combining DLP and anonymization, healthcare organizations can
- Protect patient privacy.
- Reduce Risk
- Still gain valuable insights from data

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Project Section - 2

Cloud-Based Secure Architecture.

1. **Introduction**: Data stored on cloud
   - → Benifits
     - Easy access
     - Scalable
     - Cost effective
   - → Risk - cyber attack.

2. **Common cloud attacks**:-
   - Phishing
   - DDos attacks
   - Data breaches
   - Misconfigration
   - Malware

3. **Secure artitecture Design**
   a) SDN (Software defined networking):
   → Centralized control of network
   Advantages:-
   - Better traffic control
   - Quick threat response
   - Flexible security.

   b) VLANs (virtual LANs):
   → Divide network into parts
   Example:-

VLAN 1	→ Doctors
VLAN 2	→ Admin Staff
VLAN 3	→ Patients

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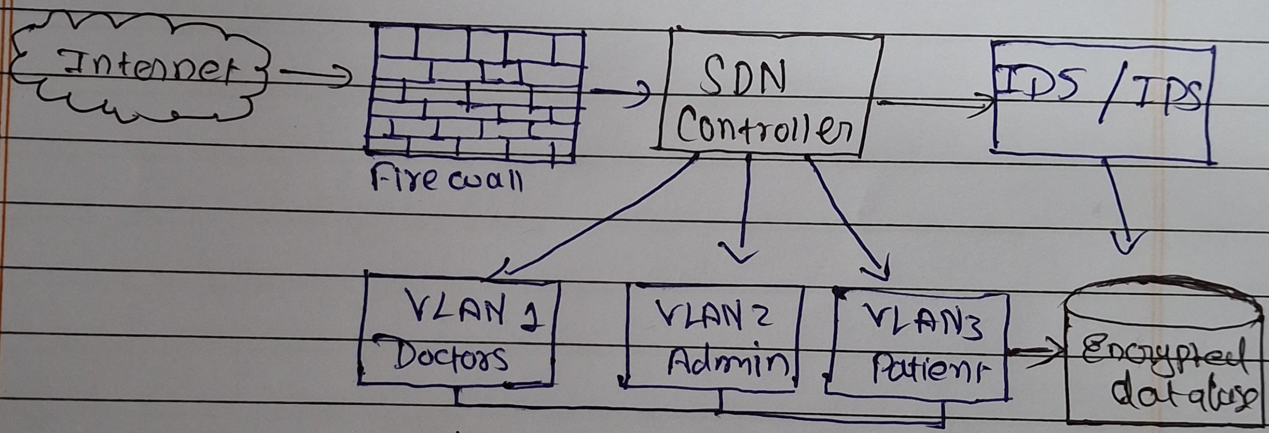
Bonifits:-

- Limit access
- Stops attack spread

c) IDS / IPS:-

- IDS - (Intrusion detection system) → detect threats
- IPS - (Intrusion prevention system) → prevents threats

4. Secure architecture design (Explanation):-



All data encrypted & monitored.

5. Additional Security measures:-

- End-to-end encryption
- Regular vulnerability scanning
- Zero trust architecture
- Backup & disaster recovery

6. Conclusion:- A cloud-based Healthcare system became secure when:-

- SDN provides control
- VLANs provides isolation
- IPS/IDS provides protection

Together ⇒ Secure cloud system.